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Columbia Laboratory for Unconventional Electronics

Capsense Capacitive Proximity Sensing Grippers for Dexterous Robotic Hand

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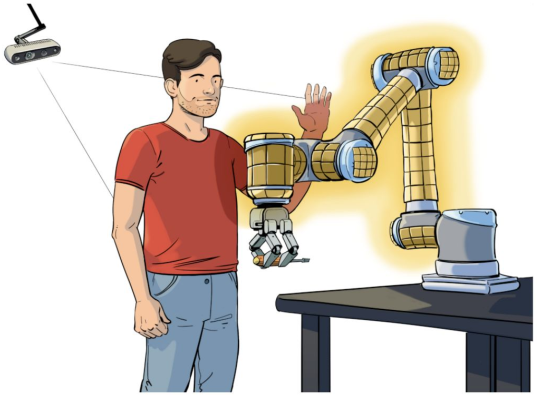
Introduction

- Current robotics perception relies on sight and touch leaving a blind spot as the hand approaches an object
    - Proximity sensing fills this gap between vision and touch

- CLUE & ROAM are developing a highly dexterous robotic hand with tactile sensors but no camera

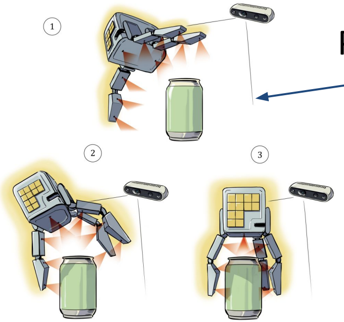
- This project integrates capacitive proximity sensing to enable pre-contact awareness and human-robot interaction



Goal

Short term goal

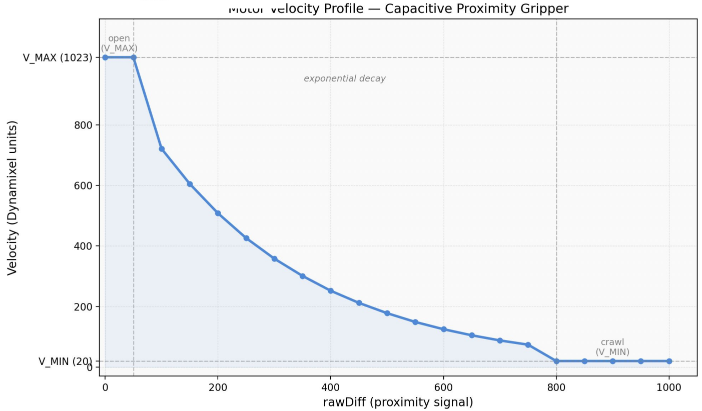
- Optimize grippers to track a conductive object
- Optimize gripper speed to be faster further away and slow down closer to the object (exponential decay)



Pre-shaping to an object

Long term goal

- Prove the concept of capacitive sensing and implement capacitive sensing in the dexterous robotic hand
- 2D/3D mapping of object using proximity values
- Object detection based on dielectric constant
- Identifying object fill level (water cup)



Conduction

Two types of conduction

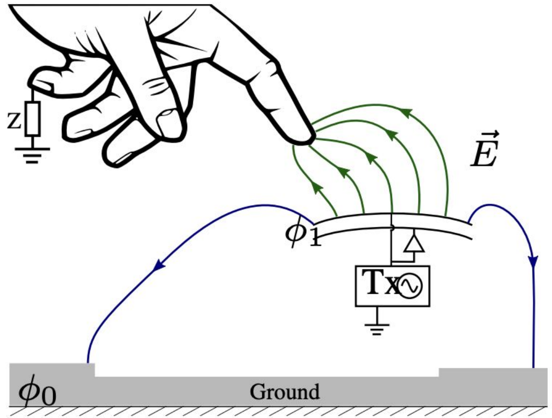
- Ionic Conductors (Hand/cup of water)
  - Lower dielectric constant
- Metallic Conductors (Metals)
  - Higher dielectric constant

$$C = \frac{\kappa \epsilon_0 A}{d}$$

Capacitance:
K = dielectric constant
ϵ0 = permittivity of free space
A = area
D = distance between capacitors

Self-Capacitance

- Single wire with wire being conductor and gnd as other conductor.
- Conductive objects placed in dielectric area between the two conductors
    - Creates two smaller capacitors in series -> distance decreases -> total capacitance increases
- Omnidirectional electric fields from both wires-> contaminates each other’s readings
- Motor noise, voltage supply ripples, EMI all affect sensor reading
- Gripper movement shifts baseline value and distance from PCB ground affecting sensor reading



Mutual-Capacitance

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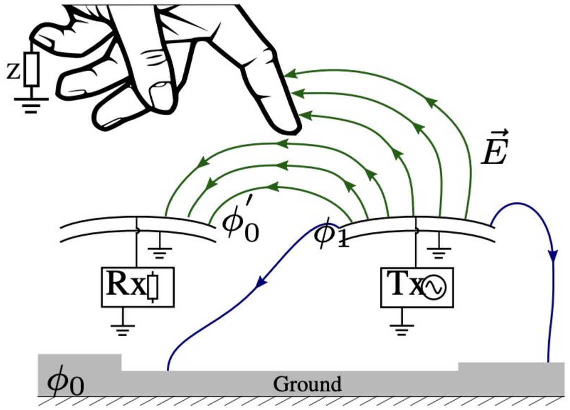
- Two wires one TX one RX
- Measures difference between two coupled electrodes
- Noise is received by both RX & TX sensors -> difference remains the same -> Less noise
- Smaller sensing area -> better for determining object position
- Independent and local sensing region -> no contaminated readings
- Baseline remains constant between RX & TX wires



Hardware

PSOC CY8C4045AZI-S413

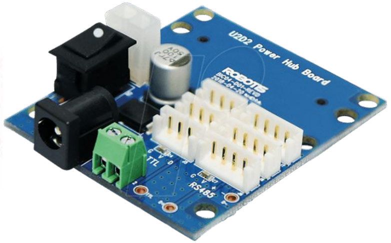
- Previous: CY8C4014LQI-421
  - Slower clock speed, worse sensing, troublesome to communicate UART
- Cypress microcontroller programmed with the MiniProg4 programmer/debugger
- Built-in Capsense proximity sensors

Dynamixel XC430-T150BBT

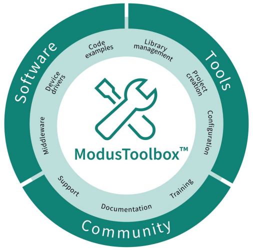
- Standard motors in robotics
- Built-in encoders that reports positioning, velocity, load, temp, and voltage
- Proprietary SDK

U2D2 USB Adaptor and U2D2 Power Hub

- Dynamixel communication from laptop using adaptor
- Power hub separates logic power (5V) from motor power (12V)



Software



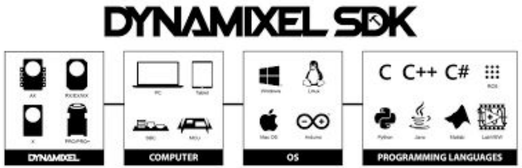
Modus Toolbox

- Proprietary PSOC software suite for programming/debugging
  - Eclipse IDE - edit code
  - Device Configurator - assign pins, establish communication protocol, clock configurations
  - Capsense Configurator - set sensing options, set threshold, other tuning
  - Modus Toolbox Programmer - programs and power chip
  - Capsense Tuner - graph and map sensor values

Dynamixel

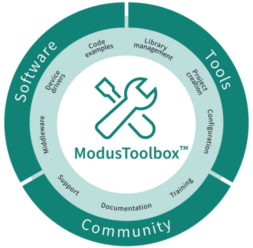
- Dynamixel_sdk for motor control
- Pyserial library for parsing UART serial communication

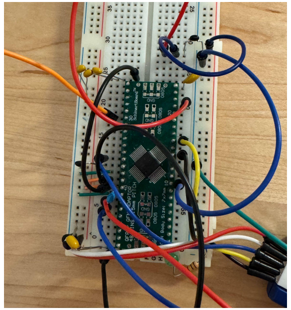
PY SERIAL

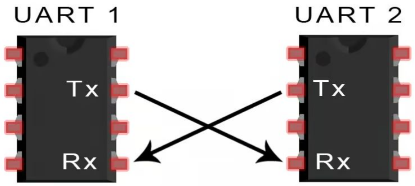


Flowchart

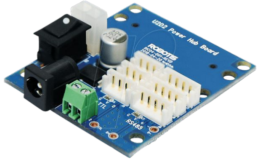
Proprietary PSOC software: IDE, Capsense Configurator, Programmer, Device Configurator

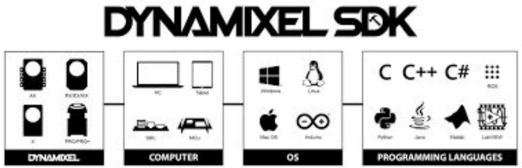


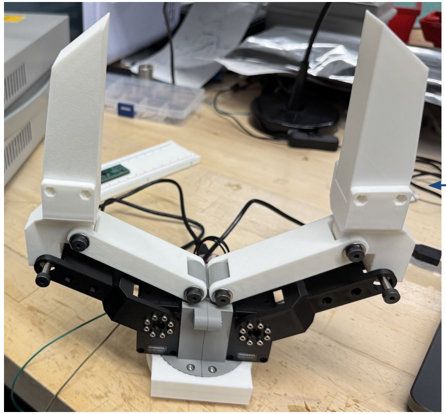




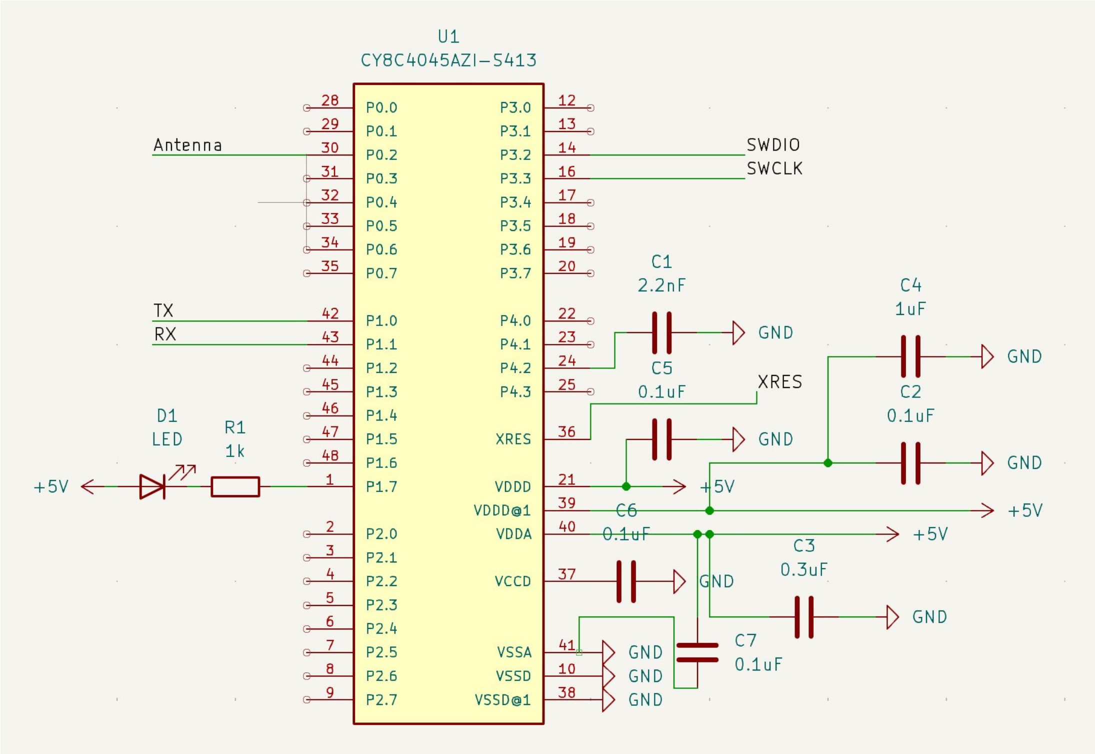
PY SERIAL





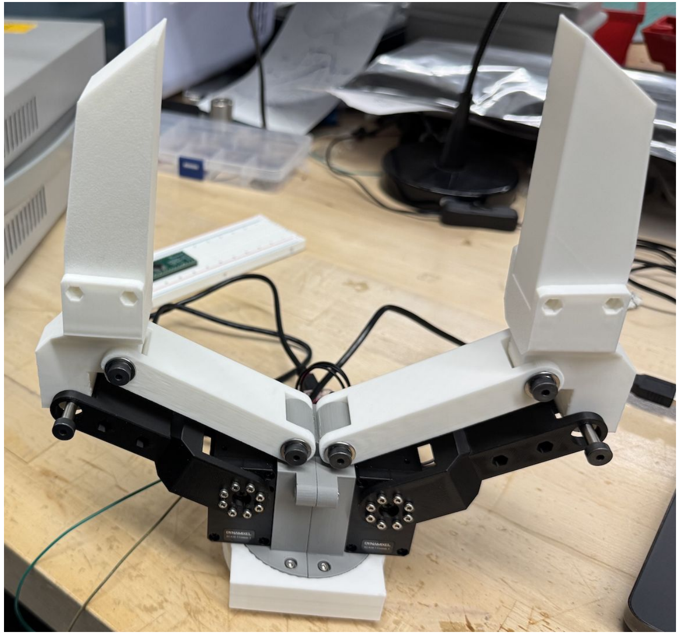


Schematic


U1
CY8C4045AZI-S413
P0.0
P0.1
P0.2
P0.3
P0.4
P0.5
P0.6
P0.7
P1.0
P1.1
P1.2
P1.3
P1.4
P1.5
P1.6
P1.7
P2.0
P2.1
P2.2
P2.3
P2.4
P2.5
P2.6
P2.7
P3.0
P3.1
P3.2
P3.3
P3.4
P3.5
P3.6
P3.7
P4.0
P4.1
P4.2
P4.3
XRES
VDDD
VDDD@1
VDDA
VCCD
VSSA
VSSD
VSSD@1
Antenna
TX
RX
SWDIO
SWCLK
XRES
D1
LED
R1
1k
+5V
C1
2.2nF
C4
1uF
C5
0.1uF
C2
0.1uF
C6
0.1uF
C3
0.3uF
C7
0.1uF
GND

Grippers

Based off open source project:
https://ut-hcrl.github.io/LEGATO-Gripper/



Capacitive Sensing

Serial port UART communication

```
=== DEBUG BOOT ===
Step 2: UART hardware OK
Step 3: retarget_io (printf) OK
Step 4: Starting CapSense init...
Step 4: CapSense init OK
Step 5: IRQ enabled, CapSense enabled
Step 6: Calibrating widgets (one-shot)...
Step 6: Calibration WARNING status=0x00000010
Step 6b: IDAC 30 -> 15 (sensitivity doubled)

>>> KEEP HAND AWAY FROM SENSOR WIRE <<<
Step 7: Sampling idle baseline in 3 seconds...
Step 7: Sampling now...
Step 8: Idle baseline seeded. Raw0=2508 Bsln0=2508

=== ENTERING MAIN LOOP ===
NOTE: Raw=4095 always means Cmod cap missing on pin 28, or sensor wire not connected

[scan 0] Raw:2489 Bsln:2508 RawDiff:-19 Diff:0 Stat:0x00 Brightness:0 idle

[scan 500] Raw:2495 Bsln:2508 RawDiff:-13 Diff:0 Stat:0x00 Brightness:0 idle

[scan 1000] Raw:2501 Bsln:2509 RawDiff:-8 Diff:0 Stat:0x00 Brightness:0 idle

[scan 1500] Raw:2521 Bsln:2510 RawDiff:11 Diff:0 Stat:0x00 Brightness:0 idle

[scan 2000] Raw:2515 Bsln:2509 RawDiff:6 Diff:0 Stat:0x00 Brightness:0 idle

[scan 2500] Raw:2508 Bsln:2513 RawDiff:-5 Diff:0 Stat:0x00 Brightness:0 idle

[scan 3000] Raw:2528 Bsln:2526 RawDiff:2 Diff:0 Stat:0x00 Brightness:0 idle

[scan 3500] Raw:2551 Bsln:2556 RawDiff:-5 Diff:0 Stat:0x00 Brightness:0 idle

[scan 4000] Raw:2585 Bsln:2580 RawDiff:5 Diff:0 Stat:0x00 Brightness:0 idle

[scan 4500] Raw:2595 Bsln:2591 RawDiff:4 Diff:0 Stat:0x00 Brightness:0 idle

[scan 5000] Raw:2615 Bsln:2599 RawDiff:16 Diff:0 Stat:0x00 Brightness:0 idle

[scan 5500] Raw:2600 Bsln:2613 RawDiff:-13 Diff:0 Stat:0x00 Brightness:0 idle

[scan 6000] Raw:2613 Bsln:2614 RawDiff:-1 Diff:0 Stat:0x00 Brightness:0 idle

[scan 6500] Raw:2633 Bsln:2626 RawDiff:7 Diff:0 Stat:0x00 Brightness:0 idle

[scan 7000] Raw:2624 Bsln:2638 RawDiff:-14 Diff:0 Stat:0x00 Brightness:0 idle

[scan 7500] Raw:2796 Bsln:2643 RawDiff:153 Diff:153 Stat:0x01 Brightness:3 DETECTED

[scan 8000] Raw:2705 Bsln:2645 RawDiff:60 Diff:60 Stat:0x00 Brightness:0 idle

[scan 8500] Raw:2613 Bsln:2634 RawDiff:-21 Diff:0 Stat:0x00 Brightness:0 idle

[scan 9000] Raw:2626 Bsln:2629 RawDiff:-3 Diff:0 Stat:0x00 Brightness:0 idle

[scan 9500] Raw:2611 Bsln:2636 RawDiff:-25 Diff:0 Stat:0x00 Brightness:0 idle

[scan 10000] Raw:2654 Bsln:2650 RawDiff:4 Diff:0 Stat:0x00 Brightness:0 idle

[scan 10500] Raw:2708 Bsln:2671 RawDiff:37 Diff:37 Stat:0x00 Brightness:0 idle

[scan 11000] Raw:2749 Bsln:2674 RawDiff:75 Diff:75 Stat:0x00 Brightness:0 idle

[scan 11500] Raw:2798 Bsln:2674 RawDiff:124 Diff:124 Stat:0x00 Brightness:2 idle

[scan 12000] Raw:2819 Bsln:2674 RawDiff:145 Diff:145 Stat:0x01 Brightness:3 DETECTED

[scan 12500] Raw:2795 Bsln:2674 RawDiff:121 Diff:121 Stat:0x01 Brightness:2 DETECTED

[scan 13000] Raw:2823 Bsln:2674 RawDiff:149 Diff:149 Stat:0x01 Brightness:3 DETECTED
```

Scan Chart - Mutual Capacitance

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The MCU has a 12-bit ADC convertor so raw value ranges from 0 to 4095

$$Raw\ Count\ = \frac{C_S * V_{supply} * NumConversion}{C_{mod} * IDAC}$$

Cs = sensor capacitance (pF)

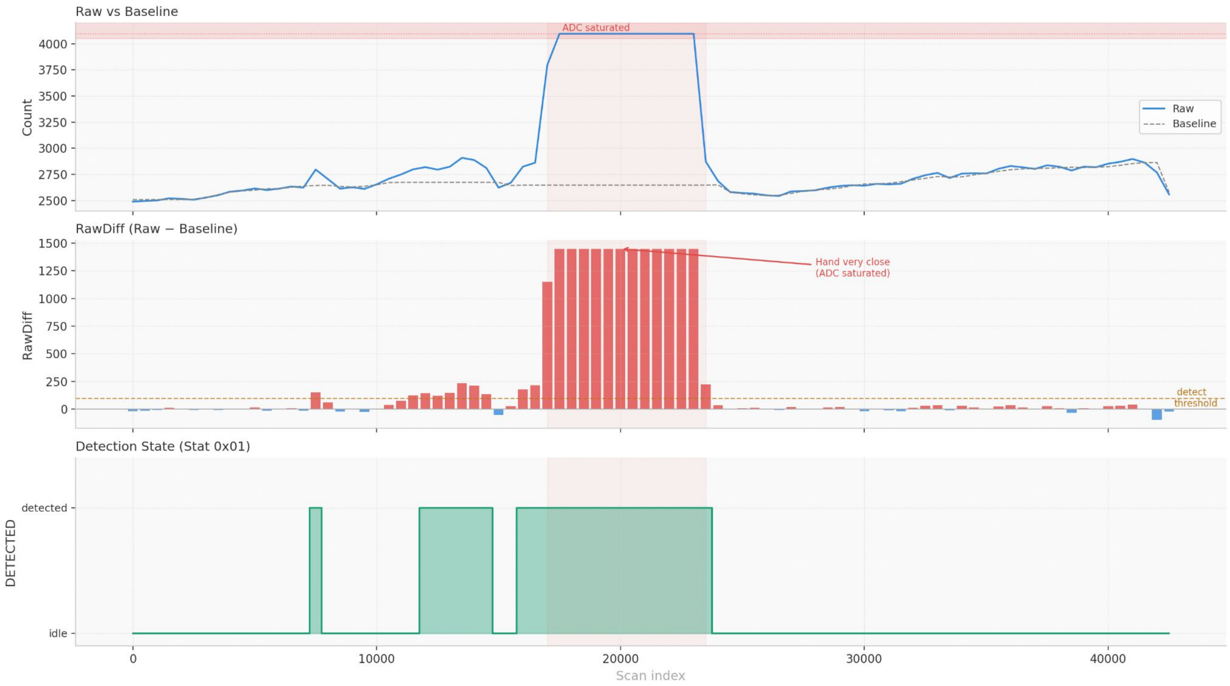
Vsupply = 5V

NumConversion = 12 bits

4096 conversions

Cmod = 2.2nF

IDAC = Current Digital to Analog Convertor



Future Steps

Next Steps:

- Currently self-capacitance is set-up, next configure two self-capacitance sensors for two gripper claws
- Two mutual capacitance setup is feasible and can be tested after the two self-capacitance setup
- Tune self-capacitance to detect ionic conductors for further away
- Integrate sensors onto gripper
- Python control script
    - Pyserial library to parse UART serial communication data
    - Motor velocity mapping to proximity capacitive values

Future Goals:

- Custom PCB for microcontroller, capacitors, resistors & LEDs
- 2D/3D mapping of objects using proximity sensing

Citations

Citations:

S. Escaida Navarro, S. Muhlbacher-Karrer, H. Alagi, H. Zangl, K. Koyama, B. Hein, C. Duriez, and J. R. Smith, "Proximity perception in human-centered robotics: A survey on sensing systems and applications," arXiv, Aug. 2021. [Online]. Available: https://arxiv.org/abs/2108.07206

S. Zodey and S. K. Pradhan, "MATLAB toolbox for kinematic analysis and simulation of dexterous robotic grippers," Procedia Engineering, vol. 97, pp. 1886–1895, 2014. doi: 10.1016/j.proeng.2014.12.342